Segment or loop? A homotopy-type-theoretic overview of the dispute over Corollary 3.12 of inter-universal Teichmüller theory

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Abstract

The dispute over Corollary 3.12 of inter-universal Teichmüller theory (IUT) III has been conducted in many vocabularies: the identification and the labelling of isomorphic copies; monodromy; segment, loop and universal cover; multiradiality and the properties IPL, SHE and APT; logical AND and logical OR; descent; commutativity up to indeterminacies; the size of the blurring; and so on. This paper shows that these become precise propositions in one language, that of homotopy type theory (HoTT)—types, paths, transport, truncation, families over graphs—and that the location of the dispute is then fixed at a single point.

There are five main results. (A) Identifying isomorphic copies along a single path loses no information. (B) If two stages are joined by two paths, a family over them is equivalent to a family over a loop, and the composite of the two paths (the monodromy) is a complete invariant that does not change however one re-identifies. For a value to be single-valued on the loop, it must be fixed by the monodromy; if the monodromy is nontrivial, the only allowed value is 0. (C) Corollary 3.12 is equivalent to a "lax 2-cell" between the two paths: a single inequality saying that the blurring by the indeterminacies absorbs the discrepancy of the monodromy, and this inequality itself has the shape of an upper bound for a height. Together with the upper bound for the blurring in IV, it gives a height bound. (D) If the blurring is taken along the whole loop, the hull becomes trivial and the inequality says nothing. (E) The properties IPL, SHE and APT are one-dimensional data (paths and functoriality) and are compatible with any size of the blurring; hence neither Corollary 3.12 nor its negation follows from them. When the same framework is applied one level down, to presented reals and Booij's locators, there is no monodromy, and the blurring of the possible images is uniformly bounded; the difference is a difference of h-level (Section 7).

(A) is the formal content of the simplification of Scholze–Stix. (B) is the formal content of the conditional statements (AD) and (IUAD), about which Mochizuki wrote that there is no dispute, and of Mochizuki's own explanation by segment, loop and universal cover. (C) and (E) show that what remains in dispute is not a question of logic but the truth of one inequality. None of the results is new mathematically; the contribution of this paper lies in the formulation that places known mathematics in one framework, and in the correspondences (§1.5). The quantitative core has been checked by machine in Lean 4. Nothing is claimed about the truth of Corollary 3.12 or of the abc conjecture.

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1 Introduction

1.1 The vocabularies of the dispute

Around Step (xi) of the proof of Corollary 3.12, each party has used its own vocabulary. The following are those treated in this paper (their sources are collected in the table of Section 6).

- Scholze–Stix [SS]: the simplification that identifies isomorphic copies. Identifying the copies of the real numbers consistently produces the monodromy j^2, so j^2 has to be omitted, which leads to an empty inequality. The blurring must be by a factor of at least $O(l^2)$.

- Mochizuki [R18]: the use of labels; the conditional statement (AD) and the absence of dispute (IUAD); the segment Γ_{seg}, the loop Γ_{loop}, the universal cover Γ_{uni} and monodromy ((LbEx1)). [IUT3]: the properties IPL, SHE and APT; two naive approaches and the "trivial multiradial representation"; the holomorphic hull of the union of possible images. [Ess]: logical AND $\wedge$ and logical OR $\vee$; descent; deletion of labels.
- LANA [LANA]: whether, after the indeterminacies are taken into account, a certain diagram commutes or a certain isomorphism is the identity; the agreement (9-1) of two isomorphisms; a re-examination of the hexagon of Scholze–Stix.
- Higuchi [Hi]: a numerical bridge does not follow from a package of qualitative properties. Dupuy–Hilado [DH]: the indeterminacies made explicit. Joshi [Jo1, Jo2]: labelled isomorphic copies; arithmetic Teichmüller spaces.

These vocabularies stand side by side without being translated into one another. The aim of this paper is to place them in one language and to show, in the form of theorems, what is agreed and what is disputed. We choose homotopy type theory as that language for three reasons. First, Mochizuki himself explains the dispute by segment, loop, universal cover and monodromy [R18, (LbEx1)], and Scholze–Stix also use the word monodromy [SS, §2.2]; in homotopy type theory these are types themselves (the interval, the circle, the universal cover). Second, in homotopy type theory an "identification" is a path, a first-class object, and a proposition cannot be written down without saying along which path one identifies. Third, univalence decides, as a theorem, when identifying isomorphic objects is harmless.

The question of this paper arose from the observation that the problem the author met in formalizing presented reals [U]—whether different presentations of the same thing are to be identified—seems to have the same shape as this dispute. Section 7 states the relation between the two as propositions.

1.2 Main results

In Section 3 we define families over graphs whose fibres are ordered real lines (what Scholze–Stix call "ordered 1-dimensional $\mathbb{R}$-vector spaces") and prove the following.

- Proposition 3.4 (segment): a family of two lines joined by one path is equivalent to a single line. The identification loses nothing.
- Propositions 3.5 and 3.7 (two paths): a family joined by two paths is equivalent to a family over a loop, and the scale $m \in \mathbb{R}_{>0}$ of the monodromy is a complete invariant.
- Proposition 3.8 (values on the loop): the values are exactly the fixed points of m; if $m \neq 1$, the only value is 0.

In Section 4 we show that the union of possible images is the smallest label-independent family (Proposition 4.2) and that deleting the labels turns a logical AND into a logical OR (Proposition 4.4). In Section 5 we put the comparison of Step (xi) into a model with two paths and show the following.

- Theorem 5.2: there is no strict 2-cell (an agreement of the two paths on the Θ-pilot).
- Theorem 5.3 and Corollary 5.4: Corollary 3.12 is equivalent to a lax 2-cell, and this is equivalent to the inequality $(1/m - 1)\,|\log q| \leq E$, which says that the blurring absorbs the discrepancy of the monodromy. Together with the upper bound $E \leq C$ for the blurring in IV, it gives a height bound.
- Theorem 5.5: if the blurring is taken over the orbit along the whole loop, the hull becomes trivial.

- Theorem 5.7: IPL, SHE and APT hold both in models in which Corollary 3.12 holds and in models in which it does not.

In Section 6 we tabulate these results and match each statement of each party with a proposition of this paper. In §6.4 we put the diagrammatic language of LANA, the indexed identifications of [UE] and the equality of this paper in one table, and state the relation between the shape of LANA's (9-1) and the shape of Corollary 3.12 as Proposition 6.1. In Section 7 we apply the same tools one level down, to presented reals and locators (Propositions 7.1, 7.5 and 7.7). There, there is no monodromy, the blurring of possible images is uniformly bounded, and the inverse of forgetting returns to a single point. The difference from IUT is a difference of h-level.

None of the above results is new mathematically. What is new is the formulation that places known mathematics in one framework, and the correspondence with the vocabulary of each party. In terms of the labels of §1.5, the propositions and theorems are labelled P or P R, and there is no proposition labelled N.

1.3 What this paper does not claim

- It makes no judgement on Corollary 3.12, on Theorem 1.10 of IV, or on the abc conjecture.
- The model of Section 5 is an abstraction of the comparison of Step (xi); how the theory constructs the blurring E, and how IV estimates it, lie outside the model. Each component of the model is given a source, but the faithfulness of the model is not itself a theorem.
- Quotations are verbatim from the original texts, with the emphasis of the originals; omissions are marked by "…", and double quotation marks inside a quotation are rendered as single ones.

1.4 Relation to earlier work

As far as the author has checked, there is no literature that reads the dispute through homotopy type theory. The geometric picture of this paper is not new, however. Segment, loop, universal cover and monodromy are in Mochizuki [R18, (LbEx1)]; monodromy and consistent identification are in Scholze–Stix [SS, §2.2]; the commutativity of diagrams and (9-1) are in LANA [LANA, §0.2, §§9–10]. The content of Theorem 5.7 (a numerical comparison does not follow from qualitative properties) has been proved by Higuchi [Hi] in a different form. The contribution of this paper is threefold: (1) placing these in one framework of homotopy type theory, stating them as propositions and proving them; (2) distinguishing "identification of fibres" from "a loop made by a second path", and separating the agreed part (Proposition 3.8, Theorems 5.2 and 5.5) from the disputed part (Theorem 5.3); (3) matching the vocabulary of each party, item by item, with propositions in this framework.

Together with this paper we publish a companion paper [UE], an educational version that writes the model of this paper without homotopy type theory, using only undergraduate mathematics (sets and maps, multiplication of positive real numbers, graphs, inequalities). In [UE], Mochizuki's elementary numerical model and the three options of Scholze–Stix are put on the same one-line inequality, using a symbol for indexed identifications.

1.5 Labels

The labels in the text mean the following.

S A statement of a source (a quotation).

X A claim of another author (a statement in a work other than the original sources).

P A proposition proved in this paper. A proposition with P only is either an elementary fact or a known fact stated in its standard form (in the latter case its source is given).

R Not new mathematically, but the notation, formulation or correspondence is that of this paper. For a proposition it is used together with P, and the source of the underlying mathematics is given as "Known mathematics".

N A theorem first proved here (as far as the author has checked). There is no proposition labelled N in this paper.

H A reading or interpretation of this paper (not a proof).

Definitions and remarks without a label are either standard (with a source) or expository.

2 Preliminaries on homotopy type theory

We work in a type theory with dependent sums Σ, dependent products Π, identity types, a univalent universe $\mathcal{U}$ and propositional truncation $\|-\|$ [RS, §2], [MO, §2]. Standard facts are cited; lemmas in the form used here are given proofs.

2.1 Types, paths and transport

For elements x, y of a type A, the elements of the identity type $x = y$ are called paths. Paths have reflexivity refl_x, composition $p \cdot q$ and inverses p^{-1} [MO, §2]. A type family $P : A \to \mathcal{U}$ and a path $p : x = y$ determine a transport $p_* : P(x) \to P(y)$. Propositions about paths can be reduced, by path induction, to the case refl [RS, §2.1].

Lemma 2.1 (Types that contract to a point)**.** P
Known mathematics: [CCHM, §3.2] (contractibility of singletons)

For a type A and $a : A$, the type $\Sigma(x : A).\,(a = x)$ is contractible, with centre (a, refl_a).

Proof. It suffices to show $(a, \mathsf{refl}_a) = (x, p)$ for every (x, p). By path induction we may assume $x \equiv a$ and $p \equiv \mathsf{refl}_a$, and then refl proves it. □

2.2 Univalence and the structure identity principle

The univalence axiom asserts that the type $X = Y$ of paths between types and the type $X \simeq Y$ of equivalences are equivalent via the natural map defined by path induction [RS, §2.1], [MO, §2]. For types with structure, the structure identity principle (SIP) says that paths between structured types coincide with structure-preserving equivalences [ACMZ, §1].

Lemma 2.2. P
Known mathematics: [ACMZ] (structure identity principle), [CCHM] (univalence)

For a type $X : \mathcal{U}$, the type $\Sigma(Y : \mathcal{U}).\,(X \simeq Y)$ is contractible. The same holds for structured types and structure-preserving equivalences.

Proof. By univalence $\Sigma(Y).\,(X \simeq Y) \simeq \Sigma(Y).\,(X = Y)$, and the right-hand side is contractible by Lemma 2.1. For structured types the same argument is made with SIP. □

Lemma 2.2 is the formal basis for what we call "identification of fibres is harmless". Isomorphic copies are equal by univalence, and nothing is lost by transporting anything along that equality.

2.3 h-levels and truncation

A type A is a proposition (a set, a groupoid) if for all $x, y : A$ the type $x = y$ is contractible (a proposition, a set). In general, an $(n+1)$-type is a type such that $x = y$ is an n-type for all x, y. The n-truncation $\|A\|_n$ is the universal way of making A an n-type, and a type is connected if $\|A\|_0$ is contractible [MO, §2]. The proposition $\exists(x : A).\, P(x)$ is defined as $\|\Sigma(x : A).\, P(x)\|_{-1}$. When proving a proposition we may use an element extracted from $\exists$.

2.4 The type of ordered real lines

Definition 2.3 (Ordered real lines). R

Known mathematics: [SS, §2.1.4]

An *ordered real line* is a set L with the structure of a one-dimensional real vector space and a total order $\leq$ such that $x \leq y \Rightarrow x + z \leq y + z$ and $x \leq y,\ 0 < r \Rightarrow rx \leq ry$. Its type is written $\mathbb{L}$. For $L, L' : \mathbb{L}$, an order-preserving linear bijection is called an *order isomorphism*; their set is written $\mathrm{OrdIso}(L, L')$, and $\mathrm{Aut}(L) := \mathrm{OrdIso}(L, L)$.

> S A global realified Frobenioid "simply amounts to a collection of ordered 1-dimensional $\mathbb{R}$-vector spaces $\mathbb{R}_v$ parametrized by the places v of k, together with a subspace" D_0 of their direct sum "of codimension 1", subject to a normalization by the local degrees; and "These considerations show that the category of global realified Frobenioids is equivalent to the category of ordered 1-dimensional $\mathbb{R}$-vector spaces."
>
> [SS, §2.1.4]

Thus $\mathbb{L}$ is the type of the "containers" of the values that log-volumes and degrees take in the dispute.

Proposition 2.4. P R

Known mathematics: [ACMZ] (structure identity principle), [MO, §2] (delooping)

(i) *For* $L, L' : \mathbb{L}$, $(L = L') \simeq \mathrm{OrdIso}(L, L')$.

(ii) $\mathrm{Aut}(L)$ *consists exactly of the maps* $x \mapsto rx$ $(r \in \mathbb{R}_{>0})$, *and the scale* $\sigma_L : \mathrm{Aut}(L) \to \mathbb{R}_{>0}$ *is a group isomorphism.*

(iii) *For* $f \in \mathrm{OrdIso}(L, L')$ *and* $g \in \mathrm{Aut}(L)$, $\sigma_{L'}(f \circ g \circ f^{-1}) = \sigma_L(g)$.

(iv) $\mathbb{L}$ *is a connected 1-type.*

Proof. (i) By the structure identity principle [ACMZ, §1], [RS, §2.1]. Equivalences preserving the vector space structure and the order (a proposition-valued relation) are exactly the order isomorphisms.

(ii) A linear endomorphism g of a one-dimensional space is a scalar multiplication: take r with $g(v) = rv$ for a nonzero v; then for $w = tv$ we get $g(w) = tg(v) = rw$. If g is bijective then $r \neq 0$, and if it preserves the order then $r > 0$ (if $r < 0$ the order is reversed). Composition corresponds to the product of scalars.

(iii) $f(g(f^{-1}(y))) = f(r\, f^{-1}(y)) = r\, y$.

(iv) L has a nonzero element (as a proposition). For $v \neq 0$, either $0 < v$ or $0 < -v$ (if $v < 0$, add $-v$ to both sides to get $0 < -v$). Taking a positive element e, the coordinate $x = te \mapsto t$ is an order isomorphism $L \cong \mathbb{R}$ (since $e > 0$, $te \leq t'e \iff t \leq t'$). Hence by (i) $\|L = \mathbb{R}\|$ holds, and $\mathbb{L}$ is connected. Moreover, by (i) the identity types are equivalent to subsets of sets of functions, hence are sets, so $\mathbb{L}$ is a 1-type. □

Remark 2.5. H By (ii)–(iv), $\mathbb{L}$ is a delooping of the group $\mathbb{R}_{>0}$ [MO, §2]. Since $\mathbb{R}_{>0}$ is commutative, the scale of an automorphism can be read as a single positive real number that depends neither on which line we are in nor on how we identified ((iii)). This is why the monodromy defined below has a meaning as a "number".

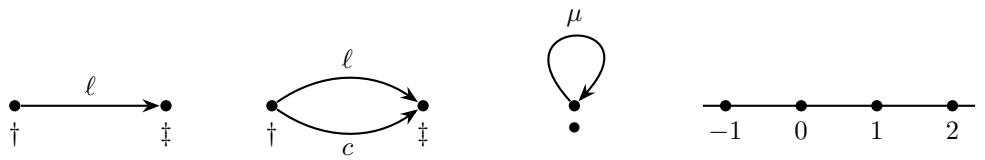
3 Families over graphs: segment, loop and universal cover

3.1 Definitions

Definition 3.1 (Graphs). R
Known mathematics: [R18, (LbEx1)]

A graph $\Gamma = (V, E, \mathrm{s}, \mathrm{t})$ consists of sets V (vertices) and E (edges) and maps $\mathrm{s}, \mathrm{t} : E \to V$. We use the following four.

- Γ_{seg}: vertices $\{\dagger, \ddagger\}$, edges $\{\ell\}$, $\ell : \dagger \to \ddagger$.
- Γ_{two}: vertices $\{\dagger, \ddagger\}$, edges $\{\ell, c\}$, both $\dagger \to \ddagger$.
- Γ_{loop}: vertex $\{\bullet\}$, edge $\{\mu\}$, $\mu : \bullet \to \bullet$.
- Γ_{uni}: vertices $\mathbb{Z}$, edges $\mathbb{Z}$, the edge n going $n \to n+1$.



Γ_{seg}: one path Γ_{two}: two paths Γ_{loop}: a loop Γ_{uni}: the universal cover

Definition 3.2 (Families). R
Known mathematics: [MO, §2]

The type of families of ordered real lines over a graph Γ is

$$\mathrm{Fam}(\Gamma) := \Sigma\big(L : V \to \mathbb{L}\big).\ \Pi(e : E).\ \big(L(\mathrm{s}\, e) = L(\mathrm{t}\, e)\big).$$

By Proposition 2.4 (i), the path on an edge e is the same thing as an order isomorphism $e_* : L(\mathrm{s}\, e) \to L(\mathrm{t}\, e)$.

Definition 3.3 (Values). R
Known mathematics: [MO, §2]

The type of *values* (sections) of a family $F = (L, p)$ is

$$\mathrm{Sec}(F) := \Sigma\big(x : \Pi(v : V).\, L(v)\big).\ \Pi(e : E).\ \big(e_*\, x(\mathrm{s}\, e) = x(\mathrm{t}\, e)\big).$$

It consists of a value at each vertex such that the values agree, whichever edge they are transported along.

Remark 3.3′. $\mathrm{Fam}(\Gamma)$ is equivalent to the type of maps into $\mathbb{L}$ from the geometric realization of Γ (the higher inductive type with the vertices as points and the edges as paths), and $\mathrm{Sec}(F)$ to the type of dependent functions over it. The realization of Γ_{loop} is the circle $\mathbb{S}^1$ [MO, §2], [CCHM]. In this paper we use Definitions 3.2 and 3.3 as they stand.

3.2 Segment: identifying fibres loses no information

Proposition 3.4. P R
Known mathematics: [CCHM, §3.2] (path induction via the contractibility of singletons)

(i) The map $\mathrm{Fam}(\Gamma_{\mathrm{seg}}) \to \mathbb{L}$, $(L, \ell) \mapsto L(\dagger)$, *is an equivalence.*

(ii) For a family $F : \mathrm{Fam}(\Gamma_{\mathrm{seg}})$, *the map* $\mathrm{Sec}(F) \to L(\dagger)$, $x \mapsto x(\dagger)$, *is an equivalence.*

(iii) Hence, for any proposition P on $\mathrm{Fam}(\Gamma_{\mathrm{seg}})$, P holds for all families if and only if it holds for the families of the form (L, L, refl_L).

Proof. (i) $\mathrm{Fam}(\Gamma_{\mathrm{seg}}) \simeq \Sigma(L^\dagger : \mathbb{L}).\, \Sigma(L^\ddagger : \mathbb{L}).\, (L^\dagger = L^\ddagger)$, and the inner part is contractible by Lemma 2.1. (ii) $\mathrm{Sec}(F) \simeq \Sigma(x^\dagger : L^\dagger).\, \Sigma(x^\ddagger : L^\ddagger).\, (\ell_* x^\dagger = x^\ddagger)$, and the inner part is contractible by Lemma 2.1. (iii) By (i), every family is equal to a family of the form (L, L, refl); transport P along that equality. □

Lean: `segment_section`

Proposition 3.4 says that identifying copies along one path changes the truth of no proposition. In this sense the simplification of Scholze–Stix ("simply identifying identical objects along the identity" [SS, §2.1], and the "obvious" isomorphisms of [SS, footnote 8]) and Mochizuki's "it is always possible to consider the situation in which one **deletes** the **labels** '†', '‡"' [Ess, Example 2.4.8 (iv-c)] are correct.

3.3 Two paths make a loop

Proposition 3.5. P R

Known mathematics: [MO, §2], [CCHM]

The map

$$\mathrm{Fam}(\Gamma_{\mathrm{two}}) \longrightarrow \Sigma(L : \mathbb{L}).\, (L = L), \qquad (L, \ell, c) \longmapsto \big(L(\dagger),\ \ell \cdot c^{-1}\big)$$

is an equivalence.

Proof. Rearrange $\mathrm{Fam}(\Gamma_{\mathrm{two}}) \simeq \Sigma(L^\dagger : \mathbb{L}).\, \Sigma((L^\ddagger, c) : \Sigma(L^\ddagger : \mathbb{L}).(L^\dagger = L^\ddagger)).\, (L^\dagger = L^\ddagger)$. By Lemma 2.1, $\Sigma(L^\ddagger).(L^\dagger = L^\ddagger)$ contracts to $(L^\dagger, \mathsf{refl})$, so the whole is equivalent to $\Sigma(L^\dagger).(L^\dagger = L^\dagger)$. That this contraction sends the remaining path ℓ to $\ell \cdot c^{-1}$ is checked by path induction on c, reducing to the case $c \equiv \mathsf{refl}$. □

Definition 3.6 (Monodromy)**.** R

Known mathematics: [R18, (LbEx1)], [SS, §2.2]

The *monodromy* of a family $F = (L, \ell, c) : \mathrm{Fam}(\Gamma_{\mathrm{two}})$ is the automorphism $c_*^{-1} \circ \ell_* \in \mathrm{Aut}(L(\dagger))$, and its scale is written $m(F) := \sigma\big(c_*^{-1} \circ \ell_*\big) \in \mathbb{R}_{>0}$.

Proposition 3.7 (The monodromy is a complete invariant)**.** P R

Known mathematics: [MO, §2] (families over a delooping and group actions)

(i) If $F = F'$ (an identification of families), then $m(F) = m(F')$. Concretely, re-identifying by order isomorphisms $\varphi : L(\dagger) \cong L'(\dagger)$ and $\psi : L(\ddagger) \cong L'(\ddagger)$, so that $\ell' = \psi \ell \varphi^{-1}$ and $c' = \psi c \varphi^{-1}$, does not change m.

(ii) The scale gives an equivalence $\|\mathrm{Fam}(\Gamma_{\mathrm{two}})\|_0 \simeq \mathbb{R}_{>0}$. That is, two families are equal (as a proposition) if and only if the scales of their monodromies are equal.

Proof. (i) m is a function on $\mathrm{Fam}(\Gamma_{\mathrm{two}})$, so it is preserved along paths. Concretely, $c'^{-1}\ell' = \varphi\,(c^{-1}\ell)\,\varphi^{-1}$, and by Proposition 2.4 (iii) the scale does not change under conjugation.

(ii) Surjectivity: for $r \in \mathbb{R}_{>0}$, the family $(\mathbb{R}, \mathbb{R}, x \mapsto rx, \mathrm{id})$ has scale r. Injectivity: by Proposition 3.5 it suffices to consider (L_1, μ_1) and (L_2, μ_2) (with $\mu_i : L_i = L_i$) such that $\sigma(\mu_1) = \sigma(\mu_2)$. By Proposition 2.4 (iv) we may take (as a proposition) a path $q : L_1 = L_2$. The transport of μ_1 along the family $L \mapsto (L = L)$ is $q^{-1} \cdot \mu_1 \cdot q$ (the transport formula in identity types [MO]), and its scale equals $\sigma(\mu_1)$ (Proposition 2.4 (iii)). Since σ_{L_2} is injective, $q^{-1}\mu_1 q = \mu_2$, and hence $(L_1, \mu_1) = (L_2, \mu_2)$. □

Lean: `monodromy_relabel`, `route_factor`

Propositions 3.5 and 3.7 say that as soon as a second path is brought in, a family over the two stages becomes a family over a loop, and that its monodromy does not disappear however one re-identifies. What the identification along the first path (Proposition 3.4) removes is only the distinction of the vertex ‡; the discrepancy between the two paths remains.

3.4 Values on the loop are fixed points

Proposition 3.8. P R

Known mathematics: [MO, §2], [R18, (LbEx1)]

Let m be the scale of the monodromy of a family $F : \mathrm{Fam}(\Gamma_{\mathrm{two}})$. Then

$$\mathrm{Sec}(F) \;\simeq\; \{\, x \in L(\dagger) \mid m\,x = x \,\}.$$

In particular, if $m \neq 1$ the only value is zero, and if $m = 1$ then $\mathrm{Sec}(F) \simeq L(\dagger)$. Moreover, "every x extends to a value" is equivalent to $m = 1$.

Proof. $\mathrm{Sec}(F) \simeq \Sigma(x^\dagger).\,\Sigma(x^\ddagger).\,(\ell_* x^\dagger = x^\ddagger) \times (c_* x^\dagger = x^\ddagger)$. Contracting $x^\ddagger$ with the second equation gives $\Sigma(x^\dagger).\,(\ell_* x^\dagger = c_* x^\dagger) \simeq \Sigma(x^\dagger).\,(c_*^{-1}\ell_*\, x^\dagger = x^\dagger)$, and by Proposition 2.4 (ii) the condition is $m x^\dagger = x^\dagger$. If $m \neq 1$, then $(m-1)x = 0$ gives $x = 0$. □

Lean: `loopSection_iff_fixed`, `loopSection_iff_zero`, `all_values_consistent_iff`

3.5 The universal cover

Proposition 3.9. P R

Known mathematics: [R18, (LbEx1)] (universal cover)

For $L : \mathbb{L}$ and $g \in \mathrm{Aut}(L)$ (of scale m), build a family over Γ_{uni} by putting L at every vertex and g on every edge. Its type of values is equivalent to L, via $x \mapsto (g^n x)_{n\in\mathbb{Z}}$. In particular, values exist for every m.

Proof. A value is a sequence $(x_n)_{n\in\mathbb{Z}}$ with $g\,x_n = x_{n+1}$, and it is determined uniquely by x_0 as $x_n = g^n x_0$ (with g^{-1} for negative n). □

Lean: `uni_edge`

3.6 Mochizuki's three graphs

> S "One considers the *('line segment') graph* Γ_{seg} … given by *two vertices* joined via a *single edge*. Write Γ_{loop} for the *('loop') graph* obtained from Γ_{seg} by identifying the *two distinct vertices* of Γ_{seg} — i.e., which may be thought of as *distinct labels* — to a *single vertex*. … situations where some sort of **'parallel transport'** or **'analytic continuation'** along the **loop** of Γ_{loop} gives rise to some sort of **nontrivial monodromy operator**. This sort of nontrivial monodromy may be **resolved** … *by working over* Γ_{seg} *as opposed to* Γ_{loop}. When, moreover, one wishes to distinguish situations that arise *prior to* and *subsequent to* multiple applications of the monodromy operator, it is natural to consider the *universal covering* Γ_{uni} … of Γ_{loop}" …
>
> "one may understand the **incompatibility** of the **Θ-link** … with **arithmetic degrees** … as a sort of **nontrivial monodromy** around the **loop** … that arises as soon as one *identifies* the *domain* and *codomain* of the Θ-link — an incompatibility … that is in fact **avoided** in IUTch … precisely by **distinguishing** *the (ring structures in the) domain and codomain of the Θ-link*, i.e., by working with 'Γ_{seg}', and then applying the **multi-radial algorithms** of IUTch to obtain an alternative way to **compute this nontrivial monodromy**"
>
> [R18, §14, (LbEx1)]

H The propositions of this paper turn this explanation into types as it stands. On Γ_{seg} the values are unconstrained (Proposition 3.4). On Γ_{loop} the values are restricted to the fixed points of the monodromy (Proposition 3.8). On Γ_{uni} there are values for every monodromy (Proposition 3.9). However, as Proposition 3.5 shows, identifying "the *domain* and *codomain* of the Θ-link" needs a second path. In our reading, the multiradial algorithm provides a second path c, different from the Θ-link. Therefore, as soon as one compares by two paths, the family becomes a family over a loop (Proposition 3.5). The monodromy is an invariant whose value is determined by what Mochizuki calls "an alternative way to **compute this nontrivial monodromy**", and it does not disappear by re-identification (Proposition 3.7). That the incompatibility is "**avoided**" in IUTch can be read as meaning that single-valuedness on the loop (Proposition 3.8) is not required. The question that then remains is treated in Section 5.

4 Possible images, logical AND and OR, and descent

Definition 4.1 (Possible images)**.** R

Known mathematics: [IUT3, Cor. 3.12] (possible images)

For a map $\pi : X \to Y$ and a family of subsets $\Phi : X \to \mathcal{P}(Z)$, the family of *possible images* is

$$\Psi_\pi\Phi(y) := \{\, z \mid \exists(x : X).\ \pi(x) = y \ \wedge\ z \in \Phi(x) \,\}.$$

Here X is the data with labels (choices of indeterminacies), Y the data with the labels forgotten, and π the forgetting.

Proposition 4.2 (Universality)**.** P R

Known mathematics: [MR] (existential quantification is left adjoint to substitution)

For $\Psi : Y \to \mathcal{P}(Z)$,

$$\big(\forall y.\ \Psi_\pi\Phi(y) \subseteq \Psi(y)\big) \iff \big(\forall x.\ \Phi(x) \subseteq \Psi(\pi x)\big).$$

That is, the union of possible images is the smallest label-independent family that contains all the labelled images.

Proof. ($\Rightarrow$) If $z \in \Phi(x)$, then $z \in \Psi_\pi\Phi(\pi x) \subseteq \Psi(\pi x)$. ($\Leftarrow$) We are to show $z \in \Psi(y)$ (a proposition) from $z \in \Psi_\pi\Phi(y)$, so we may extract from $\exists$ an x with $\pi x = y$ and $z \in \Phi(x)$. Then $z \in \Psi(\pi x)$, and transporting along $\pi x = y$ gives $z \in \Psi(y)$. □

Lean: `possibleImages_universal` (uses no axioms)

Definition 4.3 (Labelled pairs and unlabelled sets)**.** R

Known mathematics: [Ess, Example 2.4.8 (iv-c)]

For a family of two paths $F = (L, \ell, c)$ and $x \in L(\dagger)$, define the *labelled pair* $\lambda_x : \{\text{link}, \text{core}\} \to L(\ddagger)$ by $\lambda_x(\text{link}) = \ell_* x$ and $\lambda_x(\text{core}) = c_* x$, and the *unlabelled set* $U_x := \{w \mid \exists r.\ \lambda_x(r) = w\}$. The former corresponds to "along both paths it is attached in this way" ($\wedge$), the latter to "along one of the paths it is attached in this way" ($\vee$).

Proposition 4.4. P R

Known mathematics: [Ess, Example 2.4.8 (iv-c)]

(i) *U_x does not change if the roles of the two paths are interchanged.*

(ii) *If $m(F) \neq 1$ and $x \neq 0$, then $\lambda_x(\text{link}) \neq \lambda_x(\text{core})$.*

(iii) *There is no function that, looking only at the unlabelled set, returns the value attached on the side of the Θ-link. That is, there is no $G : \mathcal{P}(\mathbb{R}) \to \mathbb{R}$ with $G(U_x) = \ell_* x$ for all families and all x.*

Proof. (i) The image of a function on a two-point set does not change when the swap is composed in front. (ii) Proposition 3.8. (iii) Compare, at $x = 1$, the family $(\mathbb{R}, \mathbb{R}, x \mapsto 2x, \mathrm{id})$ and the family obtained by interchanging its two paths. By (i) they have the same $U_1 = \{1, 2\}$, but the values on the side of the Θ-link are 2 and 1, which differ. □

Lean: `unlabelled_swap`, `labelled_distinct`, `link_value_not_recoverable`

> S "it is always possible to consider the situation in which one **deletes** the **labels** '†', '‡', but only at the expense of **sacrificing** these **crucial logical AND** properties (iv-a), (iv-b), i.e., at the expense of agreeing to work under the assumption that the 'glued data' is the data associated to '†' **OR** the data associated to '‡', but *not necessarily both simultaneously*."
>
> [Ess, Example 2.4.8 (iv-c)]

H After the copies are identified, glued data appear as two paths into the same container. Proposition 4.4 says that when the labels are deleted, what remains is only the set U_x ($\vee$), and the pair λ_x ($\wedge$) does not come back.

Definition 4.5 (Descent)**.** R

Known mathematics: [Ess, §3.9]

For a map $p : A \to B$ and a construction $\Phi : A \to C$, we say that Φ *descends* to B if there is $\Psi : B \to C$ with $\Phi \circ \mathrm{id} = \Psi \circ p$ (as a pointwise identification). When B is a truncation or a quotient of A and C is a set, descending is equivalent to Φ being unchanged by the identifications of A.

> S Descent (§3.9): "a *functorial algorithm* Φ in the *original data* **descends** to a *functorial algorithm* Ψ in the *underlying data* if there exists a functorial isomorphism
>
> $$\Phi \xrightarrow{\sim} \Psi|_{\textit{original data}}$$
>
> between Φ and the *restriction* of Ψ" .... (ExtInd2) (§3.6) "may be thought of as a sort of actual **descent operation**, i.e., from data that *depends* on the specification of a member of some collection of *distinct labels* to data that is *independent* of such a label specification". (DltLb) (§3.11): the distinct labels are to be deleted "**not** via the *naive, simple-minded* approach ..., but rather via the *indirect approach* of applying **descent operations**". (FrgInv) (§3.11): "the operation of **forgetting** ... can in fact, if one allows for suitable **indeterminacies**, be **inverted**."
>
> [Ess, §3.9, §3.6, §3.11]

H Definition 4.5 is the definition of [Ess, §3.9] itself. If a label-dependent construction is identified naively, one ends up requiring values on the loop (Proposition 3.8). In descent, the values are enlarged to the set of possible images (Proposition 4.2), so this does not happen. However, the result changes according to along which identifications one descends (over the orbits of which groupoid the union is taken). This is the subject of Section 5.

5 Step (xi): two paths and 2-cells

5.1 The model

> S Corollary 3.12: write $-|\log(\underline{\underline{\Theta}})| \in \mathbb{R} \cup \{+\infty\}$ for the procession-normalized mono-analytic log-volume of the holomorphic hull of the union of the possible images of a Θ-pilot object in the multiradial representation of Theorem 3.11, which is regarded as subject to the indeterminacies (Ind1), (Ind2), (Ind3); and write $-|\log(\underline{\underline{q}})| \in \mathbb{R}$ for the procession-normalized mono-analytic log-volume of the image of a q-pilot object, which is not regarded as subject to these indeterminacies. Here $|\log(\underline{\underline{q}})| > 0$. Then "it holds that $-|\log(\underline{\underline{\Theta}})| \in \mathbb{R}$, and $-|\log(\underline{\underline{\Theta}})| \geq -|\log(\underline{\underline{q}})|$".
>
> [IUT3, Cor. 3.12]

Setup 5.1 (A model of the comparison of Step (xi))**.** R

Known mathematics: [SS, §2.1.4, §2.2], [LANA, §7.1, §10.2]

Consider the following data.

(a) Two ordered real lines $L^\dagger, L^\ddagger : \mathbb{L}$: the containers of the values that log-volumes and degrees take on the two sides of the Θ-link (the $\mathbb{R}_{\odot,\Theta}$, $\mathbb{R}_{\odot,q}$ of Scholze–Stix, the $\mathbb{R}^{\mathrm{val}}$, $\mathbb{R}^{\mathrm{ss}}$ of LANA).

(b) Two paths $\ell, c : L^\dagger = L^\ddagger$. Here ℓ is the identification induced by the Θ-link (sending the abstract Θ-pilot to the abstract q-pilot), and c is the standard identification via the shared data (the coric data).

(c) $b \in L^\dagger$ is the log-volume of the Θ-pilot, and $a \in L^\ddagger$ the log-volume of the q-pilot. IPL: $\ell_* b = a$.

(d) The coordinate of $L^\ddagger$ is chosen so that c is the identity (coric coordinates). Put $Q := |\log(\underline{\underline{q}})| > 0$ and $a = -Q$. If $m := \sigma(c_*^{-1}\ell_*)$ is the scale of the monodromy, then $c_* b = -Q/m$.

(e) A blurring $E \geq 0$: the value of the holomorphic hull of the union of the possible images of the Θ-pilot, computed through c, is $-|\log(\underline{\underline{\Theta}})| := c_* b + E$ (the excess from taking the holomorphic hull of the union of Proposition 4.2).

(f) Corollary 3.12 (model): $a \leq c_* b + E$.

X Scholze–Stix write that "There is one consistent choice of isomorphisms given by using the natural isomorphisms $\mathbb{R}_{\odot,\Theta} \cong \mathbb{R}_{\odot,q} \cong \mathbb{R}$". With these isomorphisms, however, the abstract Θ-pilot object does not encode the arithmetic degree, so scalars j^2 are to be introduced on the left (Θ-) part of the diagram, and "it is clear that this will result in the whole diagram having monodromy j^2, i.e., being inconsistent" [SS, §2.2]. LANA reproduces the hexagon of Scholze–Stix, confirms that it does not commute, and reads it as follows: "The composition $\mathbb{R}_{\odot,\Theta} \to \mathbb{R}_\Theta$ scales by a factor of $\frac{1}{\ell^*}(1^2 + 2^2 + \cdots + (\ell^*)^2)$ while the composition $\mathbb{R}_{\odot,q} \to \mathbb{R}_q$ scales by a factor of 1" [LANA, §10.2].

In this reading $m = \left(\frac{1}{l^*}\sum_{j=1}^{l^*} j^2\right)^{-1}$, where $l^* = (l-1)/2$ (written ℓ^* in [SS, LANA]), and $0 < m < 1$ if $l^* \geq 2$. The theorems below use only $0 < m < 1$ and do not depend on the value of m.

Remark 5.1′ (What lies outside the model). How the theory constructs E (the multiradial algorithm of Theorem 3.11) and how IV estimates E from above (IV, Theorem 1.10) lie outside the model. The latter is used only as the assumption $E \leq C$.

5.2 There is no strict 2-cell

Theorem 5.2. P R

Known mathematics: [R18, (AD), (IUAD)], [SS, §2.2]

In Setup 5.1, $\ell_* b = c_* b$ holds only if $m = 1$ or $b = 0$. Hence, under $0 < m < 1$, $\ell_* b \neq c_* b$: the two paths do not agree on the Θ-pilot, and there is no strict 2-cell joining the two paths.

Proof. Apply Proposition 3.8 to $x = b$; note that $b = -Q/m \neq 0$. □

Lean: `Step.no_strict_cell`

S (AD) "The correspondence that appears in the **Θ-link**

$$\{\underline{\underline{q}}^{j^2}\} \quad \mapsto \quad \underline{\underline{q}}$$

(where $j = 1, \dots, l^*$) is, when considered within/relative to a **single hol. str.**, i.e., a **single ring/scheme theory**, manifestly **incompatible/inconsistent** with computing **arithmetic degrees**, unless one considers arithmetic degrees up to the **indeterminacy** *given by possible multiplication by factors 'j^2'* (where $j = 1, \dots, l^*$) — an indeterminacy which then renders the computation *meaningless*."

(IUAD) "There is **absolutely no dispute** whatsoever (between SS and HM) with respect to the validity of the assertion (AD)."

[R18, §12, (AD), (IUAD)]

H Theorem 5.2 and Proposition 3.8 are the type-theoretic form of the first half of (AD) (incompatibility), and Theorem 5.5 is that of the second half (introducing the indeterminacy renders the computation meaningless). As Mochizuki says, there is no dispute here.

5.3 Corollary 3.12 is a lax 2-cell

Theorem 5.3. P R

Known mathematics: [SS, §2.2] (the blurring must be by a factor of at least $O(l^2)$), [LANA, §0.2] (the "wall")

In Setup 5.1 the following are equivalent.

(i) *Corollary 3.12 (model):* $a \le c_* b + E$.

(ii) *A lax 2-cell:* $\ell_* b \le c_* b + E$. *That is, the value transported along the path of the* Θ*-link does not exceed the value transported along the path of the shared data, widened by the blurring.*

(iii) $(\frac{1}{m} - 1)Q \le E$.

Proof. (i)⇔(ii) by IPL, $\ell_* b = a$. (ii)⇔(iii): since $\ell_* b = -Q$ and $c_* b = -Q/m$, we have $-Q \le -Q/m + E \iff Q/m - Q \le E \iff (1/m - 1)Q \le E$. □

Lean: `Step.cor312_iff_lax`, `Step.cor312_iff_blur`

Corollary 5.4. P R

Known mathematics: [IUT4, Thm. 1.10], [Alien, Introduction]

In Setup 5.1, if $E \le C$, then Corollary 3.12 (model) implies $Q \le C/(1/m - 1)$. If $E = 0$ (no blurring), then Corollary 3.12 (model) does not hold.

Proof. From Theorem 5.3 (iii) and $1/m - 1 > 0$. If $E = 0$, the left-hand side of (iii) is positive, so (iii) fails. □

Lean: `Step.height_bound`, `Step.cor312_needs_blur`

Theorem 5.3 is the precise form of Scholze–Stix's statement that "the blurring must be by a factor of at least $\mathrm{O}(\ell^2)$" [SS, §2.2]. In multiplicative terms, the absolute value of the hull must be at most m times that of the image, that is, it must have shrunk by the factor $1/m = \frac{1}{l^*} \sum j^2 = O(l^2)$. Corollary 5.4 shows that whether this is "useless" or "the content of the theory" hinges exactly on $E \le C$, that is, on the height bound itself. Neither evaluation follows from logic alone.

Remark 5.4′ (A consistent identification is not a necessary condition). H The comparison of Theorem 5.3 requires no value on the loop (Proposition 3.8); it requires only one inequality. Hence the general principle "a meaningful comparison requires a consistent identification of the copies of the real numbers" does not hold. What remains is only the size of the blurring.

5.4 A hull along the loop says nothing

Theorem 5.5. P R

Known mathematics: [IUT3, Remark 3.11.1 (vii), (viii)]

In Setup 5.1, suppose that a value $H \in \mathbb{R}$ of the hull (in coric coordinates) bounds from above the whole orbit of the Θ-pilot under the monodromy, that is, $m^n\, c_* b \le H$ for all $n \in \mathbb{N}$. Then $H \ge 0 \ge a$. Hence Corollary 3.12 (model) holds whatever Q is, and says nothing about Q.

Proof. We have $c_* b < 0$. Suppose $H < 0$. By the Archimedean property there is an n with $m^n < (-H)/(-c_* b)$, and then $m^n c_* b > H$, a contradiction. □

Lean: `orbit_hull_nonneg`, `Step.loop_hull_trivial`

> S "one is led [at least in the absence of more sophisticated machinery!] to regard '$\underline{\underline{q}}$' as being **only well-defined up to possible confusion with '$\underline{\underline{q}}^{\lambda^n}$'**, for some indeterminate $n \in \mathbb{Z}$. That is to say, in summary, this approach gives rise to a sort of **'uninteresting/trivial multiradial representation** of '$\underline{\underline{q}}^{\lambda}$' via"
>
> $$\text{“}\{\underline{\underline{q}}^{\lambda^n}\}_{n\in\mathbb{Z}}\text{”}$$
>
> ...
>
> "this 'trivial multiradial representation' is **useless** from the point of view of applications such as the log-volume estimates given in Corollary 3.12 below ... for the following reasons: This **'trivial multiradial representation $\{\underline{\underline{q}}^{\lambda^n}\}_{n\in\mathbb{Z}}$'** is obtained by · allowing for **indeterminacies** in the **value group** portion [i.e., '$\underline{\underline{q}}^{\mathbb{Z}}$'] of the data under consideration, ..."
>
> "By contrast, the *log-volume estimates* of Corollary 3.12 below rely, in an essential way, on the fact that in the **multiradial construction algorithm** of Theorem 3.11: · the **value group** portions of the data under consideration ... are held **rigid** [i.e., are not subject to indeterminacies], · while the **unit group** portions of the data under consideration ... are subject to the **indeterminacies** (Ind1), (Ind2), (Ind3); ..."
>
> [IUT3, Remark 3.11.1 (vii), (viii)]

H Theorem 5.5 is the type-theoretic form, at the level of log-volumes, of this "trivial multiradial representation", and also that of the "empty inequality" of Scholze–Stix [SS, §2.2]. Here, too, the two sides agree.

5.5 Qualitative properties do not decide the 2-cell

Definition 5.6 (IPL, SHE and APT in the model)**.** R

Known mathematics: [IUT3, Remark 3.11.1 (iii), (iv)]

For the data of Setup 5.1:

- IPL: $\ell_* b = a$ (the Θ-link carries the Θ-pilot to the q-pilot).
- SHE: the value of the hull is a function of the shared data; that is, there is an f with $-|\log(\underline{\underline{\Theta}})| = f(c_* b)$.
- APT: the construction commutes with re-coordinatizations of the shared data; that is, $\varphi(c_* b) = c_*(\varphi\, b)$ for every order isomorphism φ.

> S IPL: the output is constructed so as to be linked, via full poly-isomorphisms of prime-strips, to the input prime-strip. SHE: the construction of the output, as well as the output itself, is expressed in "terms that are **simultaneously** valid/executable/well-defined" relative to the arithmetic holomorphic structures in both the domain and the codomain of the Θ-link. APT: "This parallel transport mechanism does **not** consist of a simple instance

of transport of *some set-theoretic region* … *via some set-theoretic function*. Rather, it consists of a **construction algorithm** …" In Step (xi), "we shall *only be concerned* with **qualitative logical aspects/consequences**" of this construction algorithm.

[IUT3, Remark 3.11.1 (iii), (iv), and Step (xi-b) of the proof of Cor. 3.12]

Theorem 5.7. P R

Known mathematics: [Hi]

For any $0 < m < 1$ and $Q > 0$ there are two data K_1, K_2 as in Setup 5.1 (with this m and Q), both satisfying IPL, SHE and APT, such that Corollary 3.12 (model) fails in K_1 and holds in K_2.

Proof. In K_1 take $E = 0$, and in K_2 take $E = (1/m - 1)Q$. IPL holds by construction. SHE holds with $f(t) = t + E$. APT holds because c is the identity in coric coordinates. Whether Corollary 3.12 holds follows from Corollary 5.4 and Theorem 5.3. □

Lean: `package_holds`, `package_does_not_decide`

H In homotopy type theory, every construction defined on the type of the shared data is automatically transported along paths (APT) and factors through the shared data (SHE). IPL is the existence of one path. That is, all three properties are one-dimensional data (paths and functoriality) and decide nothing about 2-cells between the two paths. Theorem 5.7 is the quantitative form of this. The same conclusion has been proved by Higuchi, for the passage from Step (xi-e) to Step (xi-f), in a different formalization [Hi].

S The *"possible images"* output by the multiradial algorithm "**cannot** be **directly compared** [i.e., at least in any *a priori* sense] to the *objects* in the *local* and *global Frobenioids* that appear in the *codomain* $\mathcal{F}^{\mathbb{H}\blacktriangleright\times\mu}$*-prime-strip* of the $\Theta^{\times\mu}_{\mathrm{LGP}}$-link … determined by the **arithmetic line bundle** that gives rise to the ***q*-pilot object**". This obstruction is eliminated by the $\mathcal{O}^\times$-indeterminacies acting on tensor packets of log-shells (Ob1), by the passage from arbitrary regions to hulls (Ob2), by the passage from arithmetic vector bundles to arithmetic line bundles by forming determinants (Ob3), and so on; the result "may be **compared**, in a meaningful way, to the objects determined by the *arithmetic line bundle* that gives rise to the ***q*-pilot object**".

[IUT3, Remark 3.9.5 (vii)]; see also [IUT3, Remark 3.11.1 (iv)]

Remark 5.8 (Being comparable, and the comparison holding). H The procedure of Remark 3.9.5 (vii) carries the hull of the possible images to an object of the same type as the q-pilot, and so makes the comparison "well-typed". In the language of this paper, it corresponds to the guarantee that the value in (e) of Setup 5.1 lives in $L^\ddagger$ and that the inequality of (f) can be written down. That the inequality holds, that is, the 2-cell, is a separate assertion. Moreover, SHE in Definition 5.6 is the weak reading (being a function of the shared data). If SHE is read in the strong sense that "the output is directly comparable with the structure of the codomain and contains the q-pilot", then SHE itself asserts the 2-cell and, by Theorem 5.3, contains content equivalent to the height bound. In either reading, the 2-cell does not come from logic alone.

5.6 The two readings in the same form

X In LANA's understanding, the inequality of Corollary 3.12 "is not obtained by comparing the volumes of two pilots situated on the two sides of the Θ-link" but within the arithmetic holomorphic structure on one side of it, and the proof "rests on the comparison (9-1)" between the isomorphism η^q obtained from the q-pilot and the isomorphism η^{anab}_S obtained by anabelian and Kummer-theoretic procedures (both from $\mathbb{R}^{\mathrm{val}}$ to $\mathbb{R}^{\mathrm{ss}}$), namely $\eta^q = \eta^{\mathrm{anab}}_S$ for a suitable S. Accordingly, "the non-commutativity of the diagram in Figure 7 does not, by itself, establish a defect in the intended proof strategy" (Figure 7 is the

hexagon of Scholze–Stix). On the other hand, LANA records that "the problem concentrates on the proposition of whether, after taking 1st, 2nd, and 3rd indeterminacies into account, a certain diagram is commutative, or a certain isomorphism can be said to be the identity".

[LANA, §10.2, §10.3, §9.2, §0.2]

H The hexagon of Scholze–Stix compares two paths joining the two sides of the Θ-link. LANA's (9-1) compares, on one side of the Θ-link, two paths η^q, η^{anab} from $\mathbb{R}^{\mathrm{val}}$ to $\mathbb{R}^{\mathrm{ss}}$. Both have the form of asserting a 2-cell between two paths, modulo indeterminacies (the choice of S). The propositions of Sections 3 and 5 apply as they stand if the paths (ℓ, c) are read as $(\eta^q, \eta^{\mathrm{anab}})$ and the blurring as the orbit of S. Homotopy type theory does not decide which diagram the proof goes through. What it does decide is that, in either reading, what remains is one 2-cell.

6 Overview: the vocabularies of the parties and the propositions of this paper

Legend: S what the source says; X a claim of another author; the columns "In homotopy type theory" and "This paper" give the correspondence of this paper (a reading, H). The last column lists the propositions and theorems that apply under that reading.

Vocabulary	Source	In homotopy type theory	This paper
The simplification that identifies isomorphic copies; "obvious" isomorphisms	X [SS, §2.1, footnote 8]	Identification of fibres along one path	Prop. 3.4
Deleting the labels is always possible	S [Ess, Ex. 2.4.8 (iv-c)]	The same	Prop. 3.4
Objection to the use of labels (T5); deleting labels amounts to identification or confusion (SSDLFs)	S [R18, (T5) of §4, (SSDLFs)]	Identification of vertices (with a second path, a loop)	Props. 3.5, 3.7
Γ_{seg}, Γ_{loop}, Γ_{uni}, monodromy	S [R18, (LbEx1)]	Families over a segment, over two paths ($\simeq$ a loop) and over the universal cover	Props. 3.4, 3.8, 3.9
The monodromy j^2; the non-commutativity of the hexagon	X [SS, §2.2], [LANA, §10.2]	Scale of the monodromy $m \neq 1$; absence of a strict 2-cell	Prop. 3.7, Thm. 5.2
(AD), (IUAD): incompatible within one holomorphic structure, meaningless once the indeterminacy is introduced (no dispute)	S [R18]	Values on the loop are fixed points; the hull is trivial	Prop. 3.8, Thms. 5.2, 5.5
(LbLV): the labels mark whether the pilot is identified with the log-volume of q or with that of $\{q^{j^2}\}$	S [R18]	Along which path (ℓ or c) the containers are identified	Setup 5.1 (b)
Consistent identification, omission of j^2, empty inequality	X [SS, §2.2]	The hull over the whole orbit of the loop	Thm. 5.5
The blurring must be by a factor of at least $O(l^2)$	X [SS, §2.2]	The blurring needed for a lax 2-cell	Thm. 5.3, Cor. 5.4
Commutativity of diagrams, identity of isomorphisms, (9-1)	X [LANA, §0.2, §9.2]	A 2-cell (modulo indeterminacies)	Thms. 5.2, 5.3, §5.6
The trivial multiradial representation $\{q^{\lambda^n}\}$	S [IUT3, Rem. 3.11.1 (vii)]	The orbit of the loop	Thm. 5.5

Vocabulary	Source	In homotopy type theory	This paper
The value-group portion is rigid; the unit-group portion is subject to the indeterminacies	S [IUT3, Rem. 3.11.1 (viii), Step (xi-b)]	The path of the Θ-link is used once; the blurring is on the unit-group side	Thm. 5.3 (not Thm. 5.5)
IPL, SHE, APT, HIS	S [IUT3, Rem. 3.11.1 (iii), (iv)]	Paths and functoriality (one-dimensional data)	Def. 5.6, Thm. 5.7
The holomorphic hull of the union of possible images	S [IUT3, Cor. 3.12]	The sum through truncation (∃) and its excess	Prop. 4.2, Setup 5.1 (e)
Descent, (ExtInd2), (DltLb)	S [Ess, §3.9, §3.6, §3.11]	Factorization through a quotient or a truncation	Def. 4.5, Prop. 4.2
Logical AND ∧ and logical OR ∨	S [Ess, Ex. 2.4.8 (iv-c)]	The labelled pair and the unlabelled set	Prop. 4.4
(FrgInv): forgetting can be reversed if indeterminacies are allowed	S [Ess, §3.11]	Recovery up to an orbit, not to a point	Prop. 4.2 (H)
A numerical bridge does not follow from a qualitative package	X [Hi]	1-cells do not decide 2-cells	Thm. 5.7
(Ind1), (Ind2) are group actions; (Ind3) is a map on power sets	X [DH, §1, §4.6]	Automorphisms of the shared data (a loop), and laxness by inclusion	Setup 5.1 (e)
Labelled isomorphic copies; arithmetic Teichmüller spaces	X [Jo1, §1], [Jo2, §1]	Distinguished points over the same shared data	Section 3 (H)

6.1 What is agreed

The two paths do not agree strictly (Theorem 5.2). If one makes the values single-valued on the loop, the only value is 0, and if one takes the hull over the whole orbit of the loop, the inequality says nothing (Proposition 3.8, Theorem 5.5). This is the content of (AD) and (IUAD), and, as Mochizuki writes, there is no dispute about it between Scholze–Stix and Mochizuki. Moreover, identifying copies along one path loses nothing (Proposition 3.4). This is the content of the simplification of Scholze–Stix, and Mochizuki also writes that it is always possible to consider situations in which the labels are deleted [Ess, Example 2.4.8 (iv-c)].

6.2 What is disputed

What remains is the single point of whether a lax 2-cell between the two paths holds, that is, whether $(1/m-1)Q \leq E$ holds (Theorem 5.3). Scholze–Stix, looking at the size of the blurring this needs (a factor of at least $O(l^2)$), regard the inequality as useless. Mochizuki's side holds that the multiradial algorithm provides this comparison. LANA's (9-1) states the same point as the agreement of two paths on one side. This inequality itself has the shape of a height bound ($Q \leq E/(1/m-1)$) and is not an independent lemma. By Corollary 5.4, together with the upper bound $E \leq C$ for the blurring in IV, it gives the height bound $Q \leq C/(1/m-1)$.

6.3 What logic alone does not decide

The properties IPL, SHE and APT (in the sense of Definition 5.6) are compatible with any size of the blurring (Theorem 5.7). Hence, if Step (xi) uses only "qualitative logical aspects/consequences" of these properties, then Corollary 3.12 does not follow from them, and neither does its negation. If Corollary 3.12 holds, then what provides the 2-cell must be something other than these. The candidates are the size of the values that the comparison procedure of Remark 3.9.5 (vii) actually provides, or the quantitative content of the algorithm of Theorem 3.11 (the theta functions and their values). The latter is what Remark 3.11.1 (iv) calls "hidden internal structures (HIS)", and it is used essentially in the computation of log-volumes

in IUT IV. Homotopy type theory does not provide this content, but it shows that the dispute lies only there.

6.4 Diagrams, equality and paths: LANA's diagrams and identity types

LANA writes the point in dispute in the language of diagrams: the hexagon of Scholze–Stix (Figure 7 of LANA) does not commute; what the proof needs is the compatibility (9-1); and the "wall" concentrates on the question of whether a certain diagram commutes or whether a certain isomorphism can be said to be the identity. This paper has written the same things with the equality of identity types. The companion paper [UE] writes them with a symbol for indexed identifications. In this subsection we put the three ways of writing in one table, and state the relation between the shape of (9-1) and the shape of Corollary 3.12 as Proposition 6.1.

X "The main goal now is to relate these two kinds of isomorphisms: the one arising from q and the one arising from anabelian methods. Concretely, we wish to show that there exists some suitable S such that"

$$\eta^q = \eta_S^{\mathrm{anab}}. \tag{9-1}$$

... "(9-1) indicates that the starting point η^q belongs to an orbit (an analogue of a Teichmüller deformation) generated by the indeterminacies of the formation of η_S^{anab}."

[LANA, §9.2, §9.3]

X "the indeterminacies create ambiguities in the choice of the set S in (9-1). These ambiguities are reflected in the log-volume calculation for the hull in IUT IV [13]" ... "This description also suggests that the indeterminacies must be carefully managed so as to keep the log-volume of the hull small enough to yield a meaningful inequality." ... "To reiterate, we concur that the diagram in Figure 7 does not commute, and that repairing it would require a rescaling factor that is too large to yield a proof of the *abc* conjecture." ... "However, we, the LANA project, do not have a proof of (9-1) at this time."

[LANA, §10.4, §10.5]; [13] is IUT IV in LANA's list of references

H Since $\mathbb{L}$ is a 1-type (Proposition 2.4 (iv)), equality splits into three levels.

- *Equality of values*: $x = y$ inside one line L. Since L is a set, this is a proposition.
- *Identifications*: paths $p : L = L'$. By Proposition 2.4 (i) they are the same as order isomorphisms, and together they form a set that is an $\mathbb{R}_{>0}$-torsor.
- *Agreement of identifications*: equality $p = q$ of paths. Since $\mathbb{L}$ is a 1-type, this too is a proposition. That a diagram commutes, or that an isomorphism is the identity, is an assertion at this level.

	LANA (diagrams)	[UE] (indexed identifications)	This paper (identity types)	Level
Values	Elements of the real lines $\mathbb{R}^{\mathrm{val}}$, $\mathbb{R}^{\mathrm{ss}}$ (degrees and log-volumes)	A value $x \in \mathbb{R}_\alpha$ in a container	A point $x : L$	Proposition (equality in a set)
Identifications	Isomorphisms of based real vector spaces (η^q, η_S^{anab}, the arrows of Figure 7)	The index f and $x \approx_f y$	A path $p : L = L'$	Set (an $\mathbb{R}_{>0}$-torsor)
Agreement of identifications	Commutativity of diagrams; an isomorphism being the identity; the equality of (9-1)	Composites of indices agree (monodromy 1 around a loop)	Equality $p = q$ of paths (a strict 2-cell)	Proposition
Agreement modulo indeterminacies	(9-1) for a suitable S: η^q lies in the orbit generated by the indeterminacies	A set S of indices and $\approx_S$	$\exists(g \in G).\ \varphi = g \cdot \psi$ (Prop. 6.1 (a))	Existence ((-1)-truncation)

	LANA (diagrams)	[UE] (indexed identifications)	This paper (identity types)	Level
Comparison up to blurring	The computation of log-volumes of hulls (§10.4)	$\leq_E$	A lax 2-cell $\ell_* b \leq c_* b + E$ (Thm. 5.3)	Inequality of values

Proposition 6.1 (A strict 2-cell modulo indeterminacies, and a lax 2-cell)**.** P R

Known mathematics: [LANA, §9.2, §9.3] ((9-1)), [SS, §2.2], Theorem 5.3

Let $L, L' : \mathbb{L}$, let $\varphi, \psi : L = L'$ be paths, let $G \subseteq \mathrm{Aut}(L)$ be a subset containing id*, and let b be a negative element of L. Put the possible images $\Psi_G(b) := \{\, \psi_*(g\, b) \mid g \in G \,\}$ (Definition 4.1), the scale $m := \sigma(\psi_*^{-1} \circ \varphi_*)$, and $\Sigma := \{\sigma(g) \mid g \in G\} \subseteq \mathbb{R}_{>0}$. Consider:*

(a) a strict 2-cell modulo indeterminacies: $\varphi = g \cdot \psi$ for some $g \in G$ (that is, $\varphi_* = \psi_* \circ g_*$);

(b) membership: $\varphi_* b \in \Psi_G(b)$;

(c) a lax 2-cell: $\varphi_* b \leq \sup \Psi_G(b)$; that is, $\varphi_* b \leq \psi_* b + E$, where $E := \sup \Psi_G(b) - \psi_* b$.

Then (a) and (b) are each equivalent to $m \in \Sigma$, and (c) is equivalent to $\inf \Sigma \leq m$, with $E = |\psi_* b|\,(1 - \inf \Sigma)$. In particular (a) implies (c), but the converse does not hold.

Proof. (a): $\varphi_* = \psi_* \circ g_*$ is the same as $\psi_*^{-1} \circ \varphi_* = g_*$, and since an automorphism is determined by its scale (Proposition 2.4 (ii)), this is equivalent to $m = \sigma(g)$.

(b): $\varphi_* b = \psi_*(g\, b)$ is the same as $m\, b = \sigma(g)\, b$, and since $b \neq 0$ this is equivalent to $m = \sigma(g)$.

(c): Since ψ_* is an order-preserving linear bijection, the condition is the same as $m\, b \leq \sup\{\sigma(g)\, b \mid g \in G\}$ in L. Since $b < 0$, the right-hand side is $(\inf \Sigma)\, b$, and the condition is equivalent to $m \geq \inf \Sigma$. For the same reason $E = \psi_*\big((\inf \Sigma - 1)\, b\big) = (1 - \inf \Sigma)\, |\psi_* b|$.

The converse fails: let $0 < r < m < 1$ and let G consist of id and the automorphism of scale r. Then $\inf \Sigma = r \leq m$, so (c) holds, but $m \notin \{1, r\}$, so (a) does not hold. □

H LANA's (9-1) has the shape of (a). Take $(\varphi, \psi) = (\eta^q, \eta^{\mathrm{anab}}_{S_0})$ and let G be the automorphisms generated by changing S; then "η^q lies in the orbit of the indeterminacies" is (a), and what LANA writes next, that "the rigidified q-pilot is represented in the output regions", corresponds to (b). Inside the model, (a) and (b) are equivalent. Theorem 5.3 of this paper (Corollary 3.12 in the model) has the shape of (c), with $(\varphi, \psi) = (\ell, c)$ and G the indeterminacies that produce the blurring. That the hexagon of Scholze–Stix does not commute corresponds to (a) failing for $G = \{\mathrm{id}\}$ ($m \neq 1$; Proposition 3.7, Theorem 5.2). Repairing the commutativity by a factor of at least $O(l^2)$ corresponds to enlarging G until it reaches m.

H By Proposition 6.1, the three ways of writing pose questions of the same shape: whether the scales of the indeterminacies reach the scale m of the discrepancy between the two paths. (a) requires m to lie in Σ; (c) requires only that the infimum of Σ be at most m. If they reach it, the comparison holds, and the blurring needed is then $|\psi_* b|(1 - \inf \Sigma) \geq |\psi_* b|(1 - m)$. In Setup 5.1, $|c_* b| = Q/m$, so this is exactly the $(1/m - 1)Q$ of Theorem 5.3. When LANA writes in §10.4 that the indeterminacies must be managed carefully to keep the log-volume of the hull small, this corresponds to the condition of Corollary 5.4 that the blurring fit within the upper bound $E \leq C$ of IV. We do not claim that the scale of the discrepancy between LANA's η^q and η^{anab}_S is the same number as the $m = \big(\frac{1}{l^*} \sum j^2\big)^{-1}$ of this paper. LANA sets up (9-1) on one side of the Θ-link, and the discrepancy there lies outside the model. Homotopy type theory does not decide which diagram the proof goes through (§5.6). What it does decide is that, whichever is used, what is asked is one 2-cell and one inequality.

7 The same shape one level down: presented reals and locators

The question of this paper arose from the observation that the problem met in formalizing presented reals (real numbers that carry witnesses of convergence)—whether different presentations of the same thing are to be identified—seems to have the same shape as the IUT dispute. In this section we apply the tools of Sections 3–5 at the level one below, and show, as propositions, that the difference between the two is only a difference of h-level.

7.1 The setting, and the monodromy of sets

> S "The theorem is proved over *presented* reals—regular sequences of rationals with Bishop equality—... rather than for an abstract real-number object."
>
> [U, Abstract]

> S "In univalent mathematics, equality of real numbers can be captured by identity types directly, rather than by an imposed equivalence relation, thus avoiding the use of setoids. Preservation of equality of real numbers is automatic, but the drawback is that we are prevented from making any discrete observations of arbitrary real numbers."
>
> [Bo, Introduction]

Write P for the type of presentations and $[\![-]\!] : P \to \mathbb{R}$ for the map to the real number presented. The type $\mathbb{R}$ of real numbers is a set (a 0-type). An identification $[\![x]\!] = [\![y]\!]$ of two presentations x, y of the same real number is a proposition.

Proposition 7.1 (Sets have no monodromy). P

Known mathematics: [RS] (sets)

For two points a, b of a set A and paths $p, q : a = b$, we have $p = q$. Hence, in a family of two paths with values in a set A, the monodromy $p \cdot q^{-1}$ is trivial, and the constraint on values corresponding to Proposition 3.8 is empty.

Proof. By the definition of a set (§2.3). □

Lean: `OneLevelDown.paths_in_a_set_agree` (no axioms)

Remark 7.2 (The place of Lean). H In Lean, proofs of an equality are not distinguished [Ca, §1.2]. Proposition 7.1 holds by definition in Lean (the proof is `rfl`). That is, the equality of Lean lives exactly at this level one below. This is why, when the 1-type phenomena of Sections 2–5 were written in Lean, the isomorphisms of ordered real lines had to be made explicit as positive scalar multiplications (Appendix A).

7.2 Locators and possible images

> S "every extensional, computable map from the reals to the integers is constant". For rational numbers $q < r$, a locator "produces a specific selection of one of $q < x$ and $x < r$. In particular, the same real number can have different locators, and it is in this sense that locators are structure rather than property."
>
> [Bo, Abstract, Introduction]

Definition 7.3 (Locators). A *locator* for a real number x is a function that, for each pair of rationals $q < r$, chooses whichever of "$q < x$" and "$x < r$" is true [Bo, §3].

Definition 7.4 (Rounding by one query). R

Known mathematics: [Bo, §3]

For a locator λ, put $r_\lambda(x) := 1$ if λ chooses "$0 < x$" at the pair $(0, 1)$, and $r_\lambda(x) := 0$ if it chooses "$x < 1$". The possible roundings, with the locator forgotten, are $\Psi(x) := \{\, r_\lambda(x) \mid \lambda \text{ a locator of } x \,\}$ (the possible images of Proposition 4.2, with π the map that forgets the locator).

Proposition 7.5. P R

Known mathematics: [Bo, Introduction] (the same real number can have different locators)

Let $-1 < x < 2$.

(i) For every locator λ, $|x - r_\lambda(x)| < 1$.

(ii) $x = \frac{1}{2}$ *has a locator with* $r_\lambda(x) = 1$ *and a locator with* $r_\lambda(x) = 0$. *The rounding is not determined by the real number alone.*

(iii) Every element of $\Psi(x)$ *lies at distance less than* 1 *from* x, *and any two of them differ by at most* 1. *At* $x = \frac{1}{2}$, $\Psi(x) = \{0, 1\}$.

Proof. (i) If λ chooses "$0 < x$", then $0 < x < 2$, so $|x - 1| < 1$. If it chooses "$x < 1$", then $-1 < x < 1$, so $|x - 0| < 1$. (ii) The locator that prefers the lower option (it chooses $q < x$ whenever this is true) chooses "$0 < \frac{1}{2}$" at $(0, 1)$, and the locator that prefers the upper option (it chooses $x < r$ whenever this is true) chooses "$\frac{1}{2} < 1$". (iii) From (i), from the fact that the values lie in $\{0, 1\}$, and from (ii). □

Lean: `OneLevelDown.round01_close`, `round01_depends_on_locator`, `possibleRoundings_blur`, `possibleRoundings_half`. The two locators of (ii) were built in Lean using the decidability of the order of the real numbers (classical). Since $\frac{1}{2}$ is rational, they can in principle be built constructively.

Remark 7.6 (Contrast with the level above). H In Section 5, the blurring needed to make the comparison hold was $(1/m - 1)\,|\log q|$, which grew with $|\log q|$ (Theorem 5.3); and taking the hull over the orbit of the loop made the hull trivial (Theorem 5.5). Here there is no monodromy (Proposition 7.1), so the blurring arises only from the choice of presentation (of locator) and is bounded by 1 independently of x. One level down, descent by the union of possible images is harmless. One level up, the question is whether that descent can absorb the monodromy.

7.3 Forgetting and recovery

> S "In Lean, a statement in `Prop` is proof-irrelevant, and an arbitrary witness hidden in a `Prop`-valued existential cannot in general be eliminated into `Type` (data can nevertheless be recomputed when an independent decidable search produces a witness—Appendix B.12)". Proposition B.28: "For a presented real x, the propositional form of $0 < x$ and the tail-positivity data are interconvertible without any choice principle."
>
> [U, §7.2, Appendix B.12]

Proposition 7.7. P R

Known mathematics: [RS] (unique choice), [U, Appendix B.12]

(i) (One level down) For a decidable predicate $D : \mathbb{N} \to \mathrm{Prop}$, *the least witness can be extracted from the proposition* $\exists n.\, D(n)$ *without choice. Moreover, since the type of least witnesses is a proposition, the extracted value is unique.*

(ii) (One level up) There is no isomorphism s *of ordered real lines with* $g \circ s = s$ *for every automorphism* g. *Hence, if an identification is recovered only from the proposition that an identification exists, it is determined only up to the action of* $\mathbb{R}_{>0}$.

Proof. (i) Check $D(n)$ in the order $n = 0, 1, 2, \ldots$ and return the first n for which it holds; termination is guaranteed by $\exists$. For two least witnesses a, b, $a < b$ would contradict the minimality of b, and similarly $b < a$ is impossible, so $a = b$. (ii) Take $g = 2 \cdot \mathrm{id}$; then $2s = s$, that is, $s = 0$, which contradicts s being an isomorphism. □

Lean: `OneLevelDown.restoreLeast`, `least_witness_unique` (both without axioms), `no_invariant_identification`

Remark 7.8. H (i) has the same shape as the choice-free round trip of [U, Appendix B.12], and corresponds to the unique choice of [RS] (what exists uniquely can be extracted from a (-1)-truncation). Maietti and Rosolini, following Maietti and Sambin, require that the extensional level "should be seen as an abstraction of the intensional level according to Sambin's forget-restore principle" [MR, Introduction]. (ii) is the form, one level up, of (FrgInv) of [Ess, §3.11]: "the operation of **forgetting** ... can in fact, if one allows for suitable **indeterminacies**, be **inverted**". One level down, recovery returns to a single point; one level up, only to an orbit (a torsor).

7.4 Summary

	Presented reals (one level down)	The labels of IUT (one level up)
Container of values	$\mathbb{R}$ (a set, a 0-type)	$\mathbb{L}$ (a 1-type, the delooping of $\mathbb{R}_{>0}$)
Identification	A proposition (at most one way)	Paths (an $\mathbb{R}_{>0}$-torsor)
Monodromy	None (Prop. 7.1)	The scale m (Prop. 3.7)
Label-dependent construction	Rounding by a locator (a structure)	Data of Frobenius type; the path of the Θ-link
Blurring of possible images	Uniformly bounded by 1 (Prop. 7.5)	Must absorb $(1/m - 1)\,\lvert\log q\rvert$ (Thm. 5.3)
Inverse of forgetting	To a point, by the least witness (Prop. 7.7 (i))	To an orbit (Prop. 7.7 (ii), (FrgInv))
Shape of the problem	Structure or property (between h-levels 0 and -1)	Which identification (between h-levels 1 and 0)

H At both levels the shape of the dispute is: how to bring a label-dependent construction down to the side where the labels are forgotten. One level down, the target is a set, so there is no monodromy, and the union of possible images is uniformly small. One level up, the loop made by the two paths remains, and whether descent can absorb it remains as one inequality. Homotopy type theory treats both in one language and shows that the difference is a difference of h-level.

8 Conclusion

The vocabulary of the dispute over Corollary 3.12 can be placed in a few concepts of homotopy type theory: transport along paths, families over graphs and their values, monodromy, sums through truncation, and 2-cells. There, every logical part of the dispute is answered in the form of a theorem, and the answers agree with the documents of both parties. Identification along one path is harmless; the values on the loop are fixed points; and the hull taken over the orbit of the loop says nothing. What remains is one lax 2-cell, that is, one inequality. Its truth is not decided by qualitative properties. The inequality itself has the shape of a height bound, and together with the estimate of IV it gives a height bound. When the same framework is applied one level down, to presented reals, there is no monodromy, and the possible images of rounding by locators are uniformly small. The difference is a difference of h-level. This paper claims nothing about whether the inequality holds.

A Verification in Lean

The quantitative claims of Sections 3–5 and 7 (positive scalar multiplications as the automorphisms of ordered real lines, monodromy, values, orbits, 2-cells, the package of properties, rounding by locators, and choice-free recovery) were proved in Lean 4 (v4.30.0) with Mathlib. The claims at the level of homotopy type theory (Lemmas 2.1 and 2.2, Proposition 2.4 (i), (iv), and the equivalences of Propositions 3.4–3.7) are proved in the text and are not treated in

Lean (in Lean, proofs of an equality are not distinguished [Ca, §1.2], which is incompatible with univalence).

Text	Lean declarations	Axioms used
Prop. 3.4	`segment_section`	standard three
Prop. 3.7 (i)	`monodromy_relabel`, `route_factor`	standard three
Prop. 3.8	`loopSection_iff_fixed`, `loopSection_iff_zero`, `all_values_consistent_iff`	standard three
Prop. 3.9	`uni_edge`	standard three
Prop. 4.2	`possibleImages_universal`	none
Prop. 4.4	`unlabelled_swap`, `labelled_distinct`, `link_value_not_recoverable`	standard three
Thm. 5.2	`Step.no_strict_cell`	standard three
Thm. 5.3	`Step.cor312_iff_lax`, `Step.cor312_iff_blur`	standard three
Cor. 5.4	`Step.height_bound`, `Step.cor312_needs_blur`	standard three
Thm. 5.5	`orbit_hull_nonneg`, `Step.loop_hull_trivial`	standard three
Thm. 5.7	`package_holds`, `package_does_not_decide`	standard three
Prop. 7.1	`OneLevelDown.paths_in_a_set_agree`	none
Prop. 7.5	`OneLevelDown.round01_close`, `round01_depends_on_locator`, `possibleRoundings_blur`, `possibleRoundings_half`	standard three
Prop. 7.7 (i)	`OneLevelDown.restoreLeast`, `least_witness_unique`	none
Prop. 7.7 (ii)	`OneLevelDown.no_invariant_identification`	standard three

The "standard three" are `propext`, `Classical.choice` and `Quot.sound` (used by the construction of the real numbers). There is no `sorry` and no axiom of our own. The environment is Lean v4.30.0 with Mathlib at tag `v4.30.0` (commit `c5ea003`); Lean exits with code 0, with no errors and no warnings. The source file `SegmentOrLoop.lean`, the project files and the output of `#print axioms` are in the archive deposited with this paper.

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